

Skill Week - 2

Task 2:

Code:

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/types.h>

#include <sys/wait.h>


int main()

{

    char command[100];

    pid_t pid;


    printf("=====\n");

    printf("Enter Linux command (ls, pwd, date, whoami): ");

    scanf("%99s", command);


    pid = fork();


    if (pid < 0)

    {

        printf("Fork failed!\n");    return

1;

    }


    else if (pid == 0)

    {

        // Child Process    printf("\n----- Child

Process ----- \n");    printf("Child PID :

%d\n", getpid());    printf("Parent PID :
```

```

%d\n\n", getppid());    printf("Executing
command: %s\n\n", command);

    execlp(command, command, (char *)NULL);

    // Executes only if exec() fails    perror("Execution
failed");
    exit(1);
}

else
{
    // Parent waits for child    wait(NULL);

    printf("\n----- Parent Process ----- \n");
    printf("Parent PID : %d\n", getpid());    printf("Child
PID : %d\n\n", pid);

    printf("Child Process Finished Successfully.\n");
}

return 0;
}

```

Output:

```

avulasanjana@SanjanaReddy:~$ nano fork_exec.c
avulasanjana@SanjanaReddy:~$ gcc fork_exec.c -o fork_exec
avulasanjana@SanjanaReddy:~$ ./fork_exec
Enter a Linux command: ls

Parent Process
Parent PID = 717
Child PID = 718

Child Process
Child PID = 718
LICENSE  README.md  docs          include  process.c    src
Makefile assets    fork_exec     obj       process.c.save tests
OSSP     bin       fork_exec.c   process  scripts
Child Process Completed.

```

